One Thread Through the Spectre

A step-by-step account, starting from nothing, of why one small knot design stamped on every tile of a spectre tiling joins up into a single endless ribbon. With the loopholes we looked for, and the one theorem we use from the spectre's discoverers.

1 What we are looking at

A **tiling** is a way of covering a flat surface with shapes, edge to edge, with no gaps and no overlaps, like a bathroom floor. The **spectre** is a particular shape with 14 straight sides, all the same length, discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss. Copies of it can cover the whole endless plane. What makes it famous is that they can only do so in a pattern that *never repeats*: you can never shift the whole tiling along and have it land exactly on itself.

Take one such tiling. On every tile draw the same small design: a few short pieces of ribbon. Each piece runs from one side of the tile to another, so where two tiles meet, a piece on one side continues as a piece on the other. Across the whole tiling the pieces link up into long ribbons, and they weave over and under each other the way a Celtic knot does.

Here is the question. Do all the pieces link into *one* endless ribbon? Or into many separate ribbons? Or do some of them close up into small rings?

Plate 1 follows this page: Nine tiles with the design drawn on them.

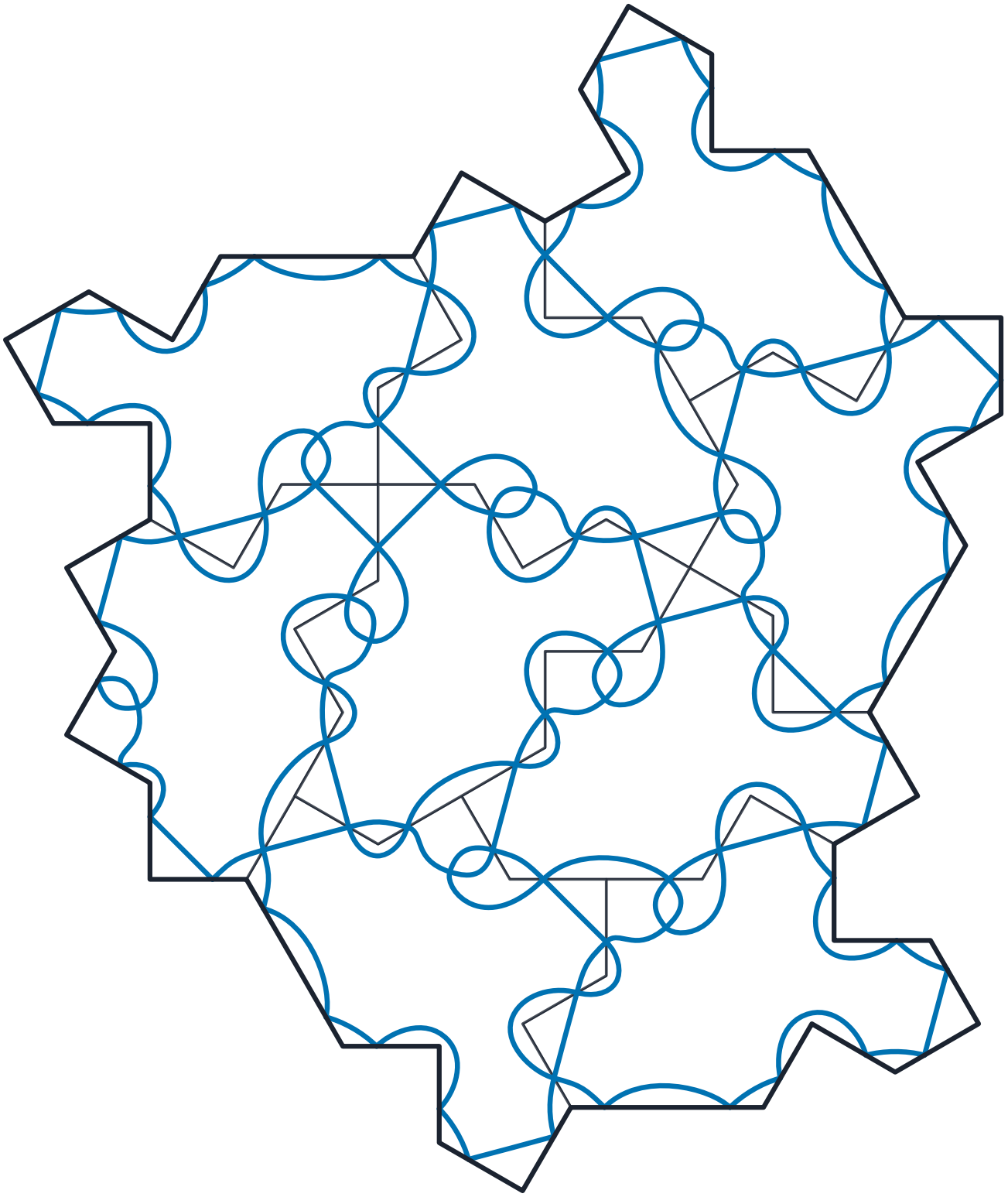
2 Why no drawing can settle it

Pick a ribbon in any drawing and follow it. Sooner or later it leaves the drawing. Whether it ever comes back, and where, is decided by tiles that are not in the picture, perhaps very far away. A finite drawing can show you a closed ring, which would answer the question with a no. It can never show you that a ribbon goes on for ever and is the only one.

If the tiling repeated, there would be a short argument that one ribbon is impossible. The spectre tiling never repeats, so that argument does not apply. But nothing hands us a proof either. What the spectre tiling does have is a kind of family tree, and the proof lives there.

3 Tiles come in families

In a spectre tiling, the tiles come in clusters of eight or nine. There are nine kinds of cluster, and we name them with Greek letters: Γ, Δ, Θ, Λ, Ξ, Π, Σ, Φ, Ψ. We call a cluster a **supertile**. The supertiles themselves come in clusters of eight or nine, again of the same nine kinds; call those level-2 supertiles. Those come in clusters too, and so on without end. Every tile belongs to a level-1 supertile, which belongs to a level-2 supertile, which belongs to a level-3 one, and so on up.



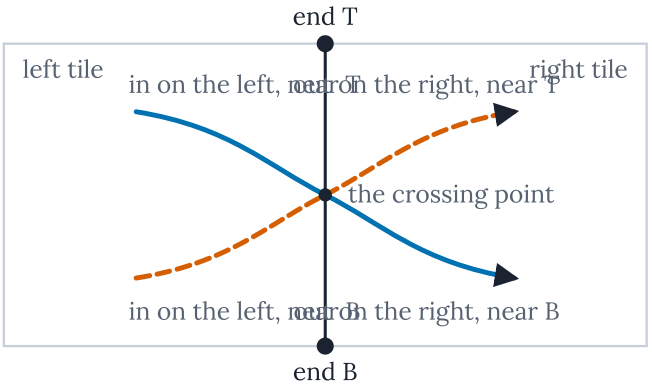
Nine tiles with the design drawn on them. Every tile carries the same drawing. The ribbon pieces run on across the tile edges. They are cut off only at the edge of this picture, where the next tiles would be. The design is called `0.1:01`. Inside the tile the ribbons cross just once. Of the simplest designs we searched, 17 different knots showed no ring in any supertile up to level 7, which is over two million tiles; this is the one whose crossing ribbons are shortest. (The over-and-under of the weave is left out of these drawings; it plays no part in the argument.)

The rule that says which kinds of supertile sit next to which, and how, is the same at every level. This endless family structure is the reason the tiling can never repeat, and it is the ladder the proof climbs.

Plate 2 follows this page: A level-2 supertile of kind Δ.

4 A tile seen from outside

Look closely at one tile edge. Where two tiles meet along it, the ribbon crosses the edge at one point (in this design 0.15 of an edge length off the midpoint, shifted alternately one way and the other round the tile, so that neighbouring tiles always agree on it), and it crosses on a slant: it arrives on one side near one end of the edge and leaves on the other side near the other end. So on each side of an edge there are two places a ribbon can be, one near each end. Call them **ports**. Crossing the edge swaps ends.



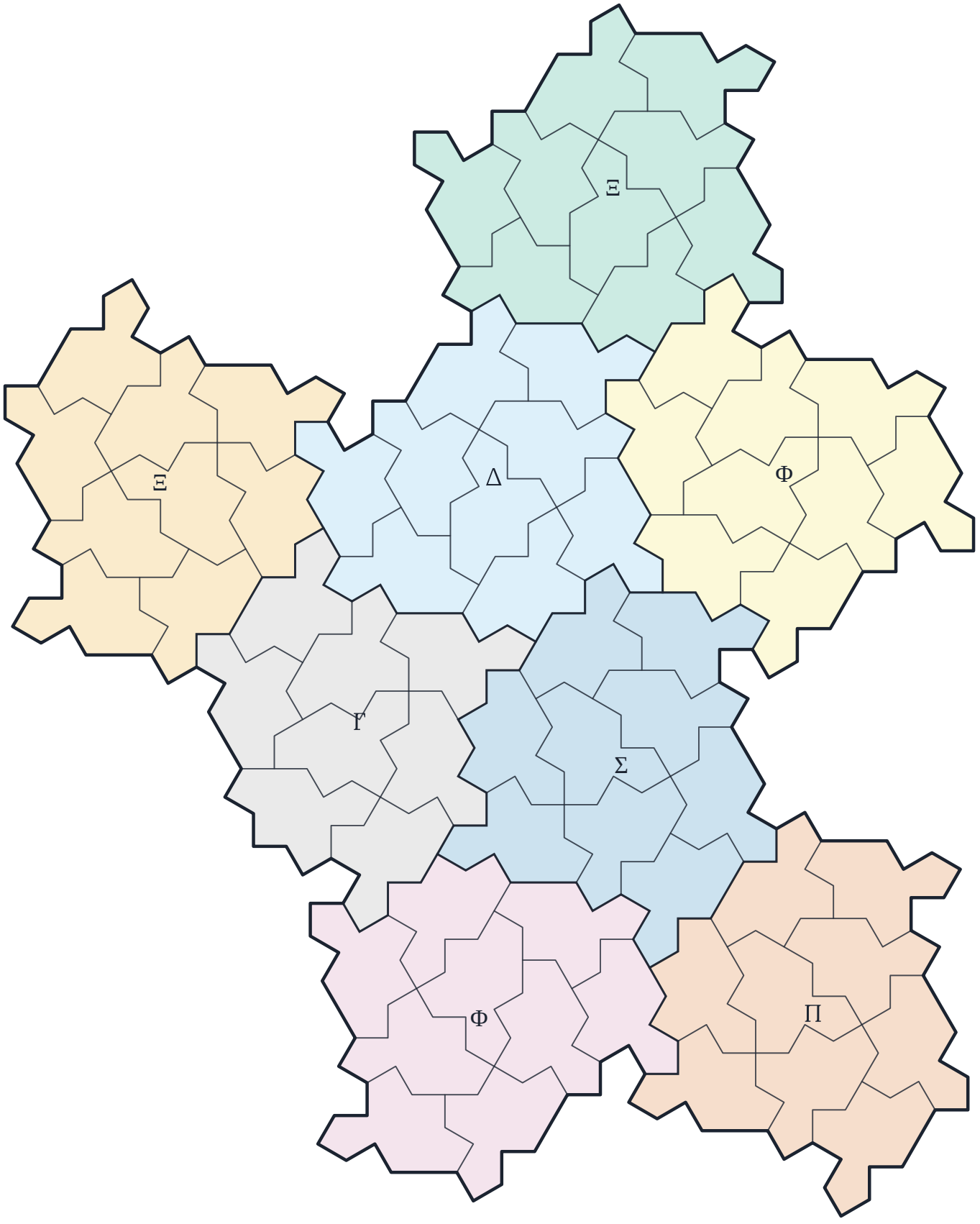
The crossing rule. A ribbon on the left side near end T becomes a ribbon on the right side near end B, and the other way round. The two ribbons cross at that point of the edge. Every edge of every tile carries exactly this.

A tile has 14 edges, so 28 ports. Inside the tile, the design joins the ports in pairs: each ribbon piece runs from one port to another. The same is true of a supertile and the ports on its rim: each piece of ribbon inside it runs from one rim port to another (or closes into a ring, if rings exist).

Here is the key observation. That pairing of ports is *all* the outside world ever learns about a supertile. Nothing outside can tell what happens inside; it only sees which port joins which. So we can find a supertile's pairing from its children's pairings alone: lay the children side by side, start at a port on the rim, follow the ribbon from port to port through the children until it comes out at the rim again, and note where. We never look inside a child. The work grows with the length of the rim, not with the number of tiles. That is why a computer can do this exactly up to level 9, where a supertile has 133 million tiles.

5 The surprise: the same simple signature at every size

Here is what the computer found for this design, for every supertile of every kind at every level from 1 to 9. At every corner of the rim except two, the ribbon that arrives on one edge turns straight round and leaves on the next edge. Call that a **hook**. At the two remaining corners the ribbons do



A level-2 supertile of kind Δ. It has 71 tiles (thin lines), grouped into eight level-1 supertiles (tinted, each labelled with its kind). The heavy line is its outer edge, which we call its **rim**. The rim is 182 tile edges long. Going up one level multiplies the number of tiles by about 8 and the length of the rim by about 4.2, so rims get very wiggly very fast.

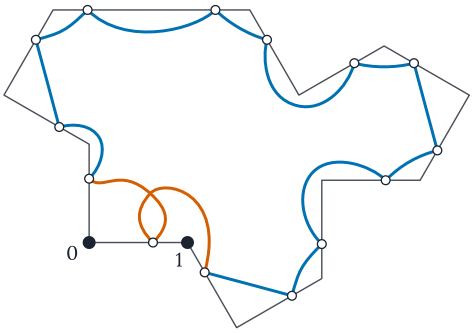
something else: they swap partners and cross each other. Call that the **crossover**. And those two corners are always in the same place: the two ends of one edge of the rim, on the supertile's Γ child. In fact the crossover is always one particular tile's own, handed up from Γ child to Γ child. Nothing closes into a ring inside. The single tile has the same signature: twelve hooks and one crossover. Its crossover is at corners 0 and 1, the two ends of edge 0. The two ribbons that meet that edge cross once inside the tile, then leave through the edges on either side.

Plate 3 follows this page: The same signature at three sizes.

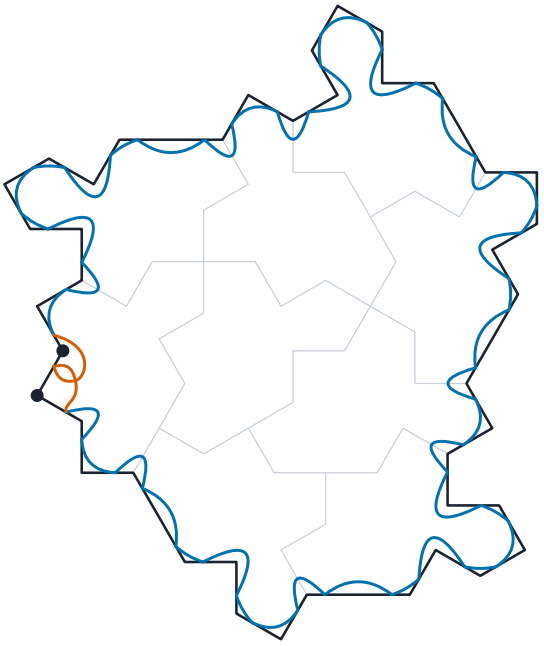
So far this is a fact found by computing, true up to level 9. On its own it says nothing about level 10, let alone the whole plane. But it has a consequence that does the heavy lifting.

6 Even or odd is all that matters

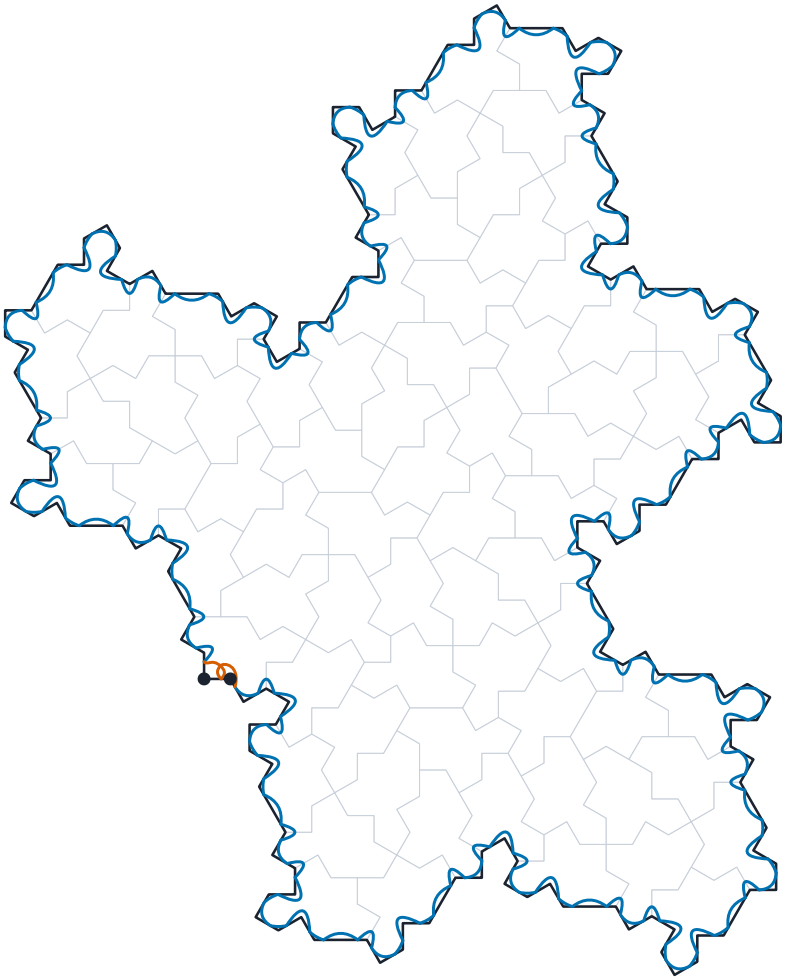
Put two supertiles side by side so that they share a run of edges. Suppose both have hooks at every corner along the run. Follow a ribbon into the run. It hooks round the first corner in one supertile, crosses the next edge on a slant into the other supertile, hooks round the next corner there, crosses back, and so on. It zigzags along the whole run, moving forward one edge at every step and changing sides at every step. After the last edge it leaves the run at the far end. If the run has an even number of edges it leaves on the side it came in on; if odd, on the other side. Nothing else about the run matters: not how long it is, not how it bends.



one tile: 14 edges, 12 hooks, one crossover

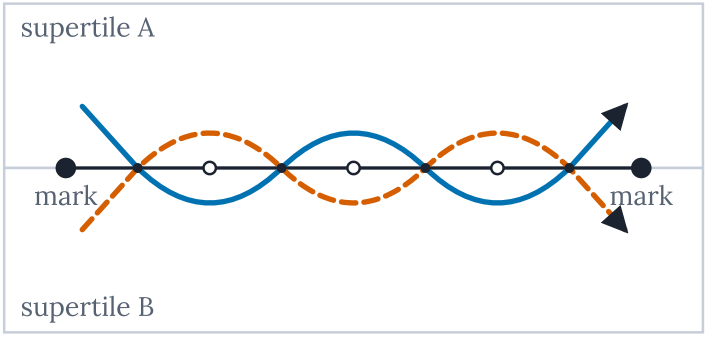


level 1: 9 tiles, 46 edges, 44 hooks, one crossover

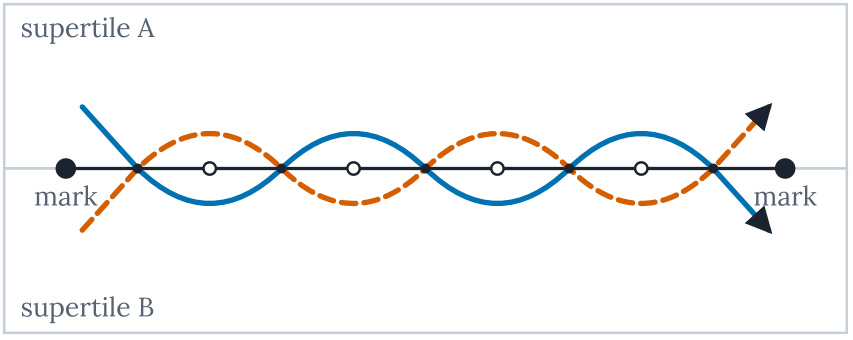


level 2: 71 tiles, 182 edges, 180 hooks, one crossover

The same signature at three sizes. Blue: hooks, one at each corner, joining the crossing points of the two edges that meet there. Orange: the crossover, at the two black corners. The tile picture is the design itself, exactly as drawn. In the bigger pictures the orange crossover is that same drawing, on the one tile that sits at the corner. The level-1 picture is the nine-tile block of figure 1 seen from outside; compare the two. The faint lines inside play no part.



4 edges (even): each ribbon leaves on the side it entered



5 edges (odd): each ribbon leaves on the other side

The zigzag. Two supertiles meet along a run of edges between two marked corners. The solid ribbon and the dashed ribbon each hook at every corner, on alternate sides, and between them they use every hook on the run. Where each one comes out depends only on whether the run has an even or an odd number of edges.

This is the step that makes the problem finite. A supertile's rim has thousands of corners and gluing two of them looks hopeless to think about. But along a shared run, all that detail boils down to one fact: even or odd.

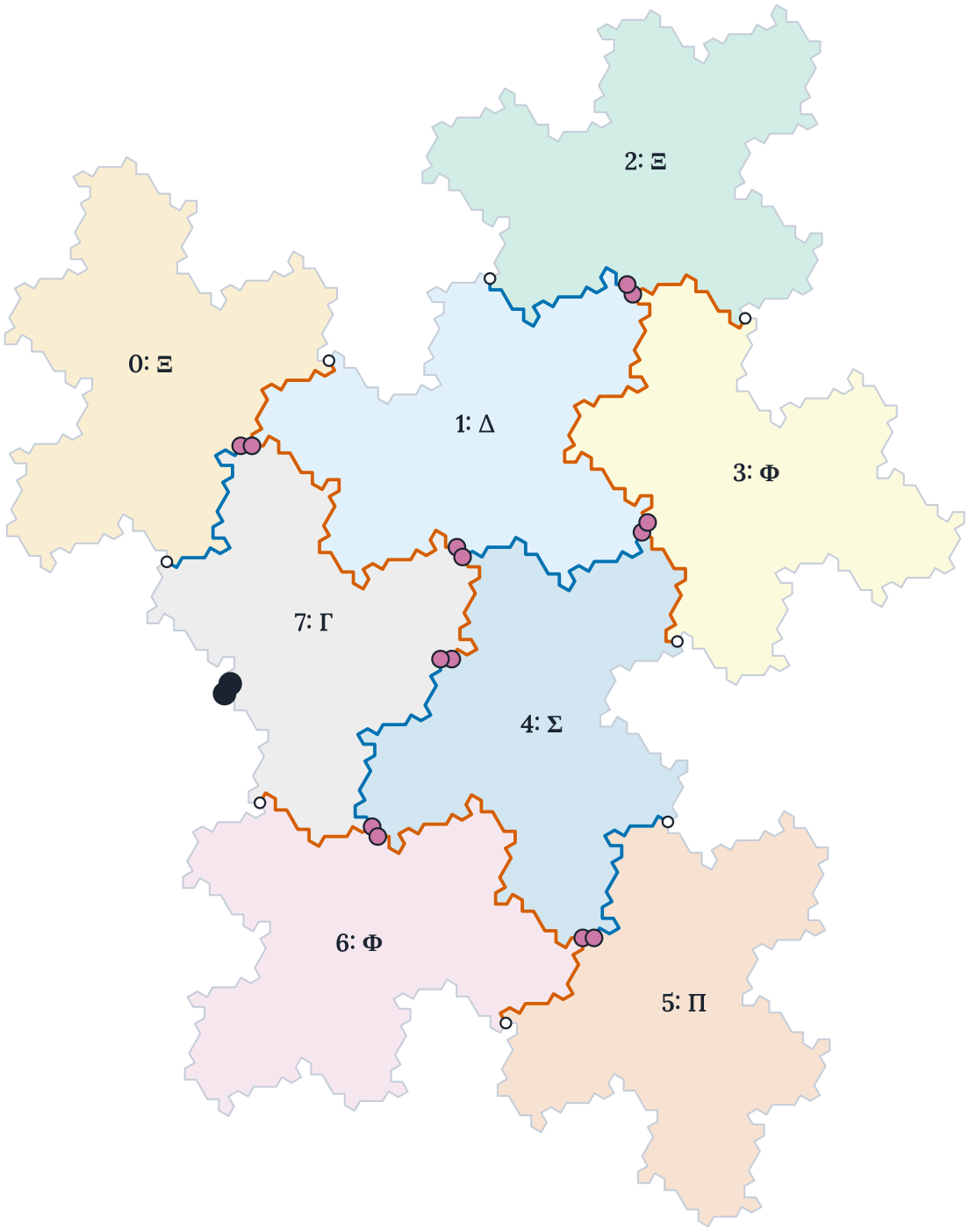
7 The small map

So, to work out a supertile's pairing from its children, we only need a small map. It shows: the children; the runs along which they touch; whether each run is even or odd; and the **marks**, which are the corners where a child's neighbour changes, plus the children's crossover corners. For a Δ supertile that is 8 children, 52 runs and 24 marks, whatever the level. Following a ribbon becomes a walk from mark to mark: a child's hook or crossover, then a run's even-or-odd rule, then the next child, until the ribbon reaches the rim. The whole walk visits at most 104 ports.

Plate 4 follows this page: The small map of a Δ supertile

8 The domino argument

We want to show the signature holds at every level, not just up to 9. The tool is the domino argument, which mathematicians call **induction**. Stand the dominoes in a line. If you can show that



The small map of a Δ supertile, drawn over a real level-3 one. Runs between children are coloured by even (blue) and odd (orange). Dots are the marks. Pink pairs are the children's crossovers; the black pair is the parent's own crossover, on the rim at the corner of its Γ child. Only this map decides the parent's pairing; the wiggly detail does not.

every falling domino knocks over the next one, and you push the first, then they all fall, however many there are.

The claim for domino n: *every level-n supertile has the signature (hooks everywhere except one crossover, at the corner of its Γ child) and has no ring inside.*

The push: levels 1 and 2 were computed outright, tile by tile. The knock-on: suppose every level-n supertile has the signature. Then the pairing of a level-(n+1) supertile is decided by its small map. The small map, read off from the real geometry, is exactly the same at levels 3, 5, 7 and 9, and exactly the same at levels 4, 6 and 8. (The odd and even ones are mirror images of each other, because the rule that builds each level includes a flip.) Run the walk on those two maps: out comes the signature, crossover at the Γ corner, no rings. So domino n knocks over domino n+1, and the chain never breaks.

Level n. Every supertile: a hook at every corner but two, one crossover at its Γ corner, no ring inside.

↓ Put the children together. Follow the ribbons from mark to mark, using only hooks, crossovers and even-or-odd. The map is the same at every level (alternating with its mirror image).

Level n+1. Every supertile: a hook at every corner but two, one crossover at its Γ corner, no ring inside.

> **Why the small map repeats for ever.** The knock-on step needs the small map to keep repeating, with its mirror image, at every level, not just the levels up to 9 that we computed. That is a fact about the tiling, not the ribbons, and it follows from what the spectre's discoverers proved, in two steps.
>
> *What the spectre paper proves.* In any tiling by the spectre, every tile lies in a unique, endless hierarchy of supertiles (Theorem 2.2). Theorems 3.1 and 4.1 turn a spectre tiling into a tiling by nine marked hexagons, and Theorem 5.1 shows those hexagons always group into supertiles that are themselves, combinatorially, mirror images of the same nine marked hexagons. So at every level the supertiles fit together exactly as the level below did: the same nine kinds, the same arrangement of children inside each kind, and the same pattern of which side of which child meets which side of which neighbour. Only the sizes change, and the picture flips at each level.
>
> *What follows from it.* Measure each side of a supertile in tile edges. A side of a level-(n+1) supertile is made of sides of its children, in that fixed pattern, so its length is a fixed sum of level-n lengths: all the lengths obey one linear rule, the same at every level. That rule still holds for the lengths taken even-or-odd. Everything the small map records (which children touch, whether each run is even or odd, where the marks fall) is read off these even-or-odd lengths and the fixed pattern. So the even-or-odd lengths at each level are determined by those at the level before, and the small map at level n+1 records all of them at level n. A process like that, with only finitely many possible states, repeats for ever once a state comes round again. The computation

shows level 4's small map equal to level 2's and level 5's equal to level 3's, so level 6 repeats level 4, level 7 repeats level 5, and so on at every level. The same argument covers the inputs to the nest in section 9, which are also even-or-odd lengths of this kind.

The spectre paper: D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss, *A chiral aperiodic monotile*, Combinatorial Theory 4(2) (2024), arXiv:2305.17743.

9 Ribbons join up

The signature says how a supertile looks from outside. It does not yet say that the pieces inside belong to one ribbon. For that, build a **nest**: a level-2 supertile, the level-3 supertile that contains it, the level-4 one that contains that, and so on for ever. Each one is child number 1 of the next. The nest grows to cover the whole plane: the inner block sits strictly inside its grandparent, with a clear gap, and the gap to the rim of the k-th ancestor multiplies by about 2.8 every level. The tiling drawn on the main page is a piece of this nest tiling.

Plate 5 follows this page: Three nested Δ supertiles.

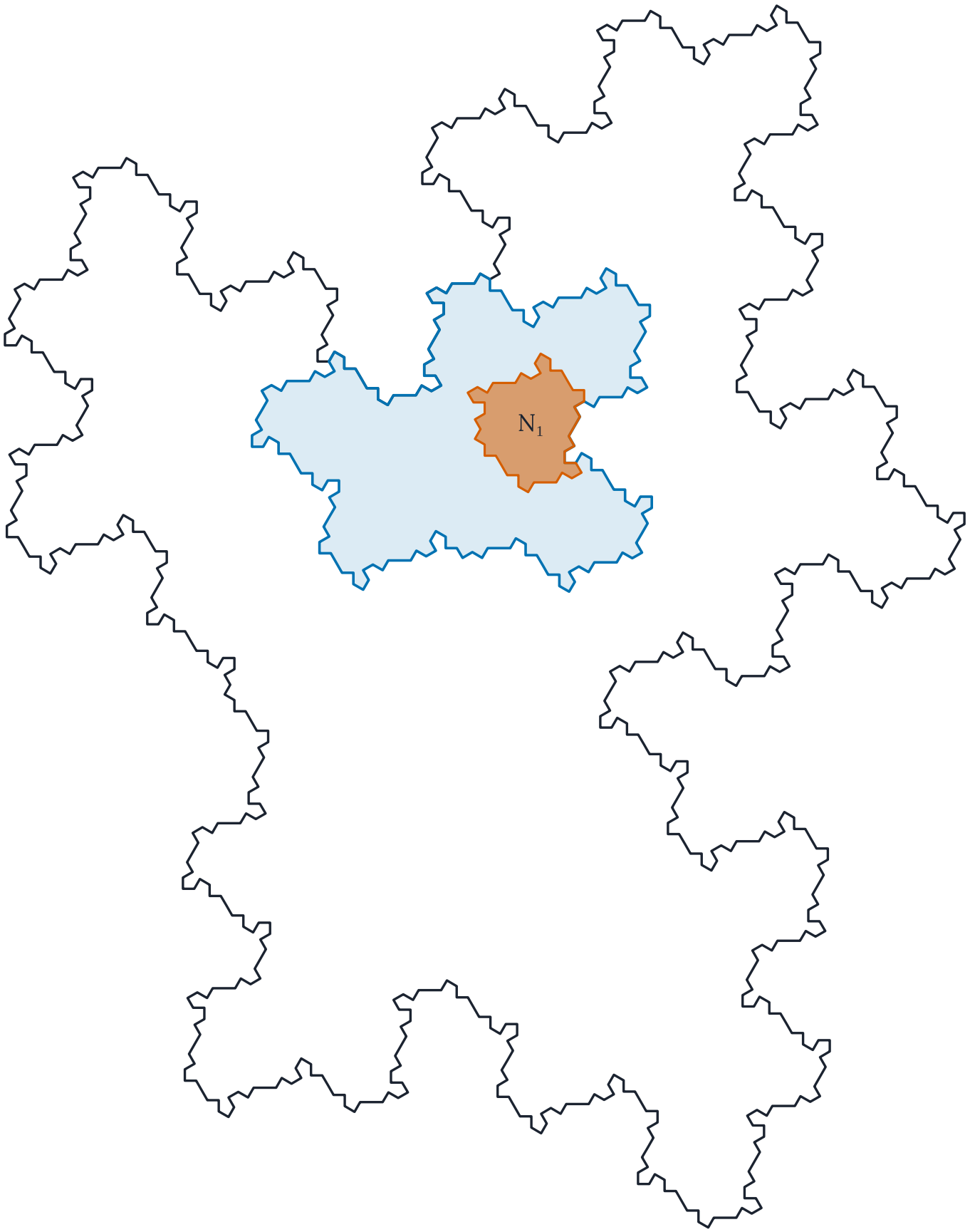
Now give every ribbon piece in the inner block its own label and carry the labels up through the same mark-to-mark walks. Along a run, the hooks are picked up alternately by the two zigzagging ribbons; at a mark, by whichever ribbon passes. Each time we go up a level, labels that sat on different ribbons find themselves on the same one. Four levels up, every label is on a single ribbon.

INNER BLOCK	PIECES	RIBBONS THEY LIE ON, 1 LEVEL UP	2 UP	3 UP	4 UP
N_1 (9 tiles)	46	10	2	2	1
N_2 (71 tiles)	182	37	2	1	1
N_3 (559 tiles)	758	146	2	2	1
N_4 (4,401 tiles)	3,198	613	2	1	1
N_5 (34,649 tiles)	13,534	2,586	2	2	1

The big numbers one level up are pieces lying along the stretch of the block's edge that is also the rim of the next block; they are gathered at the level after. This was computed two ways: exactly, following all 13,534 labels one by one, and with the small map. They agree. And the inputs to the small-map version repeat with the level in the same two-step pattern as the map itself (N_4 uses the same inputs as N_2, N_5 the same as N_3, N_6 the same as N_4), so the same holds for every block of the nest, not just these five.

10 Seeing it happen

The table in section 9 is the whole argument in numbers. Here is the same thing in pictures of the actual knot, drawn by the drawing program rather than computed from ports, so it is an



Three nested Δ supertiles. N_1 (9 tiles) inside N_2 (71 tiles) inside N_3 (559 tiles). The inner block touches the rim of its parent but sits well inside its grandparent. The nest continues outwards without end.

independent check as far as it goes. We follow the smallest block of the nest: the nine-tile block N_1, tinted orange in each picture. Its 46 ribbon pieces are the hooks and crossover of section 5.

Plate 6 follows this page: Inside the 71-tile block N2.

Plate 7 follows this page: Inside the 559-tile block N3.

Plate 8 follows this page: Inside the 4,401-tile block N4.

Plate 9 follows this page: Inside the 34,649-tile block N5.

So the pieces of the small block are gathered into 10 ribbons, then 2, then still 2, then 1, as the block it sits in grows. That is the first row of the table in section 9. The same happens to every block of the nest: the other rows are the same count for the 71-tile, 559-tile, 4,401-tile and 34,649-tile blocks.

Why no picture can finish the job: even in the last picture the single ribbon still leaves through the rim at both ends. Where it goes next is decided in the 272,791-tile block around it, and after that in the 2,147,679-tile block, and so on. A picture can only ever show a finite stage of the gathering. The domino argument is what says the gathering goes on at every level without end, and the nest argument is what turns that into "one ribbon".

11 The finish

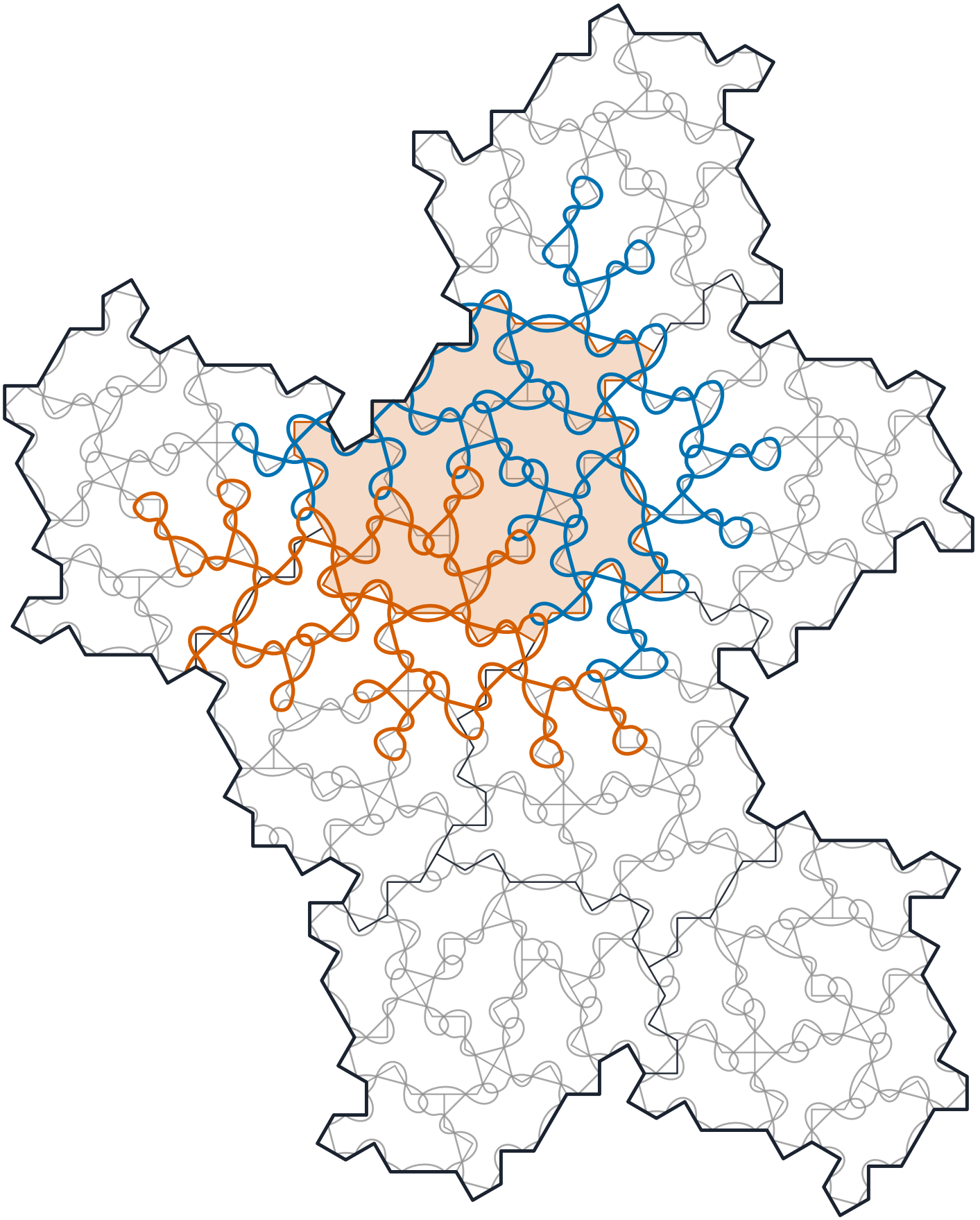
Take any two tiles anywhere in the plane. Both lie inside some block of the nest; call it N_b. Every ribbon piece in those two tiles is part of a piece of N_b, because nothing closes up inside. All pieces of N_b lie on one ribbon of N_{b+4}. So the two tiles are on the same ribbon.

And there is no ring anywhere. A ring is finite, so it sits inside some block of the nest. Take the smallest supertile that contains it. The ring is not inside any single child of that supertile, so when the supertile was assembled from its children, the ring would have been found. None ever was: at every assembly of every kind at every level, the count is zero.

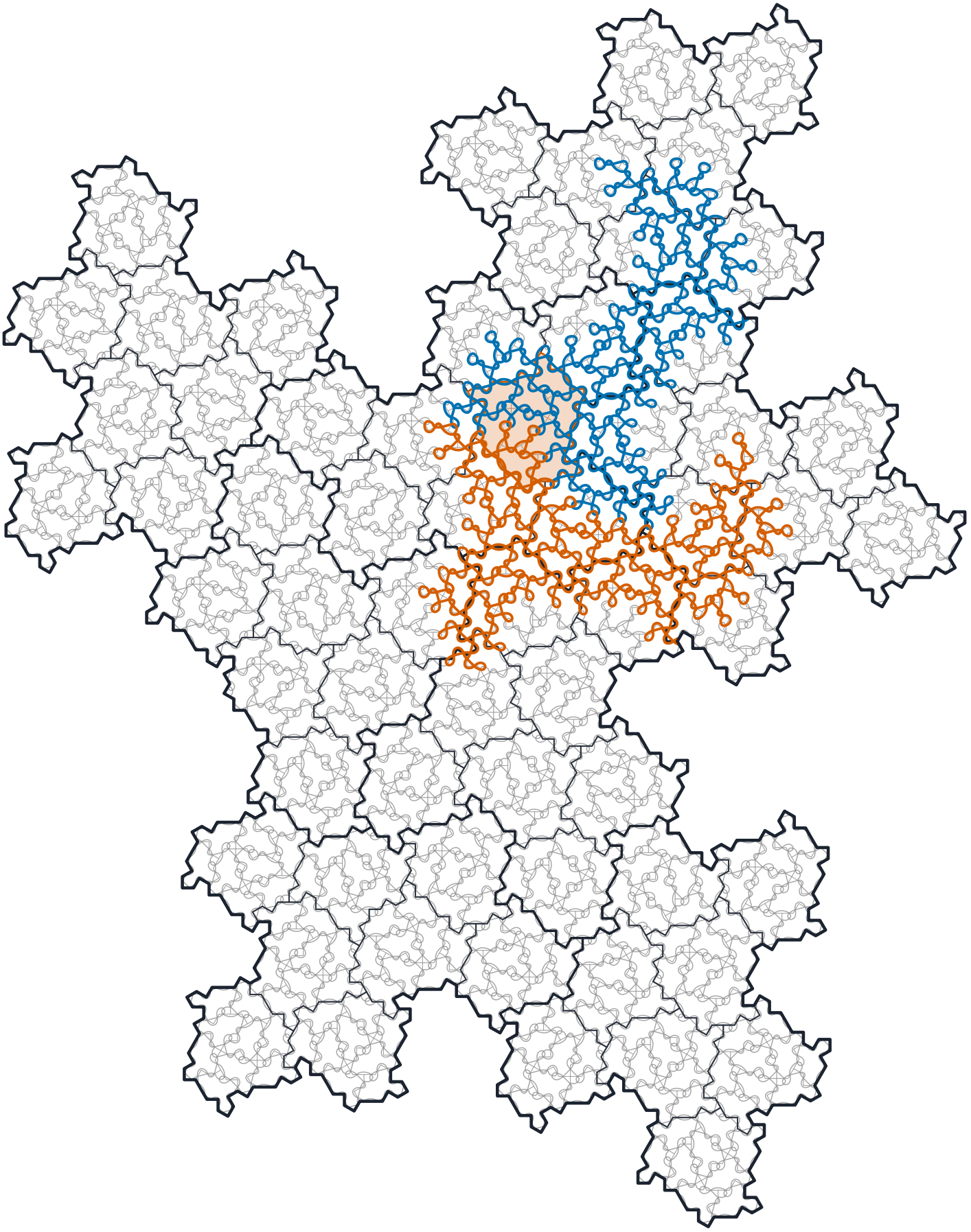
So there is one endless ribbon, and it passes through every tile. Because the tiling never repeats, neither does the ribbon.

12 Loopholes we looked for

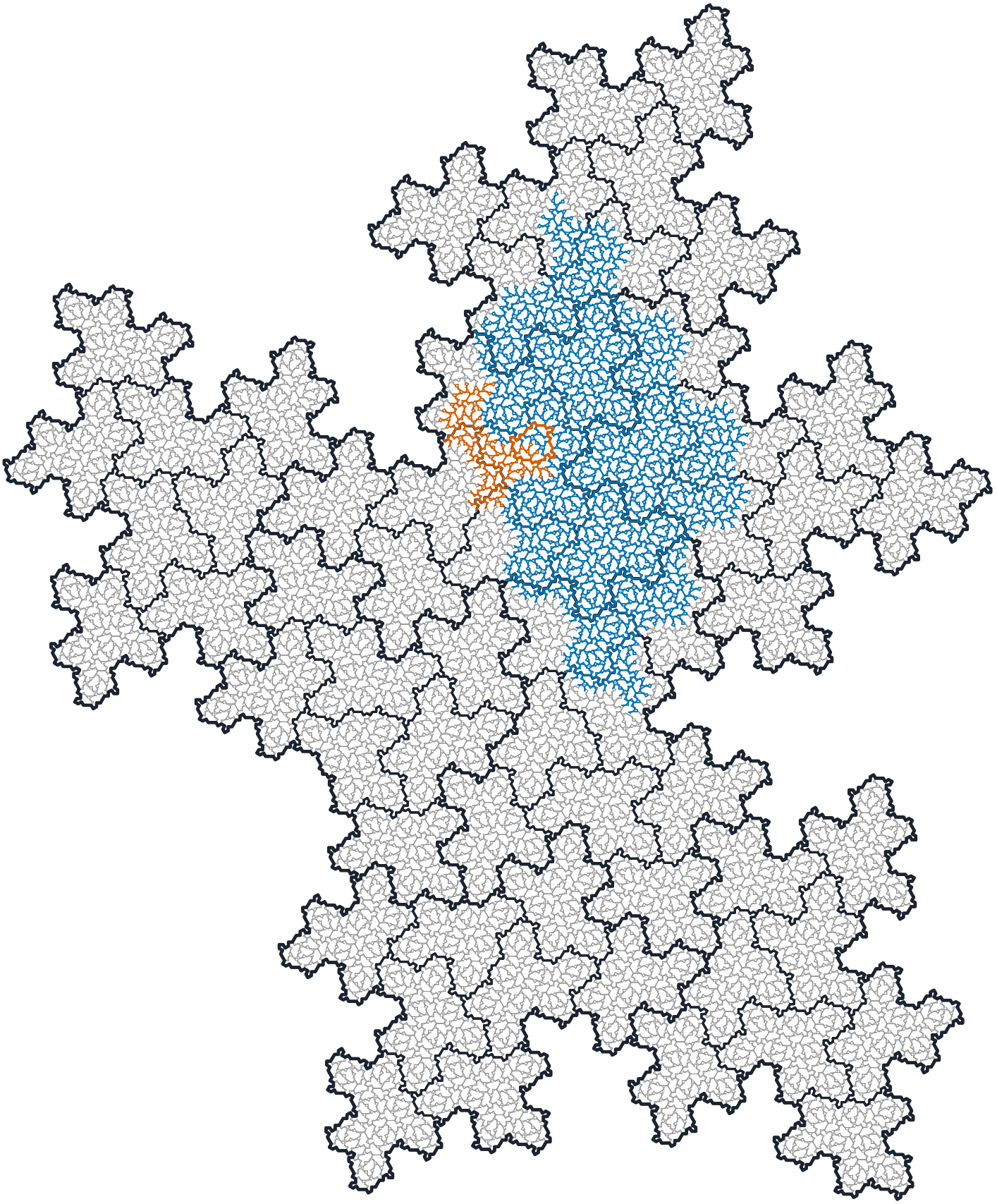
A proof like this leans on a computer, so the places it could go wrong are the places where the computer might be answering a different question from the one we asked. Here is each one and how it was closed.



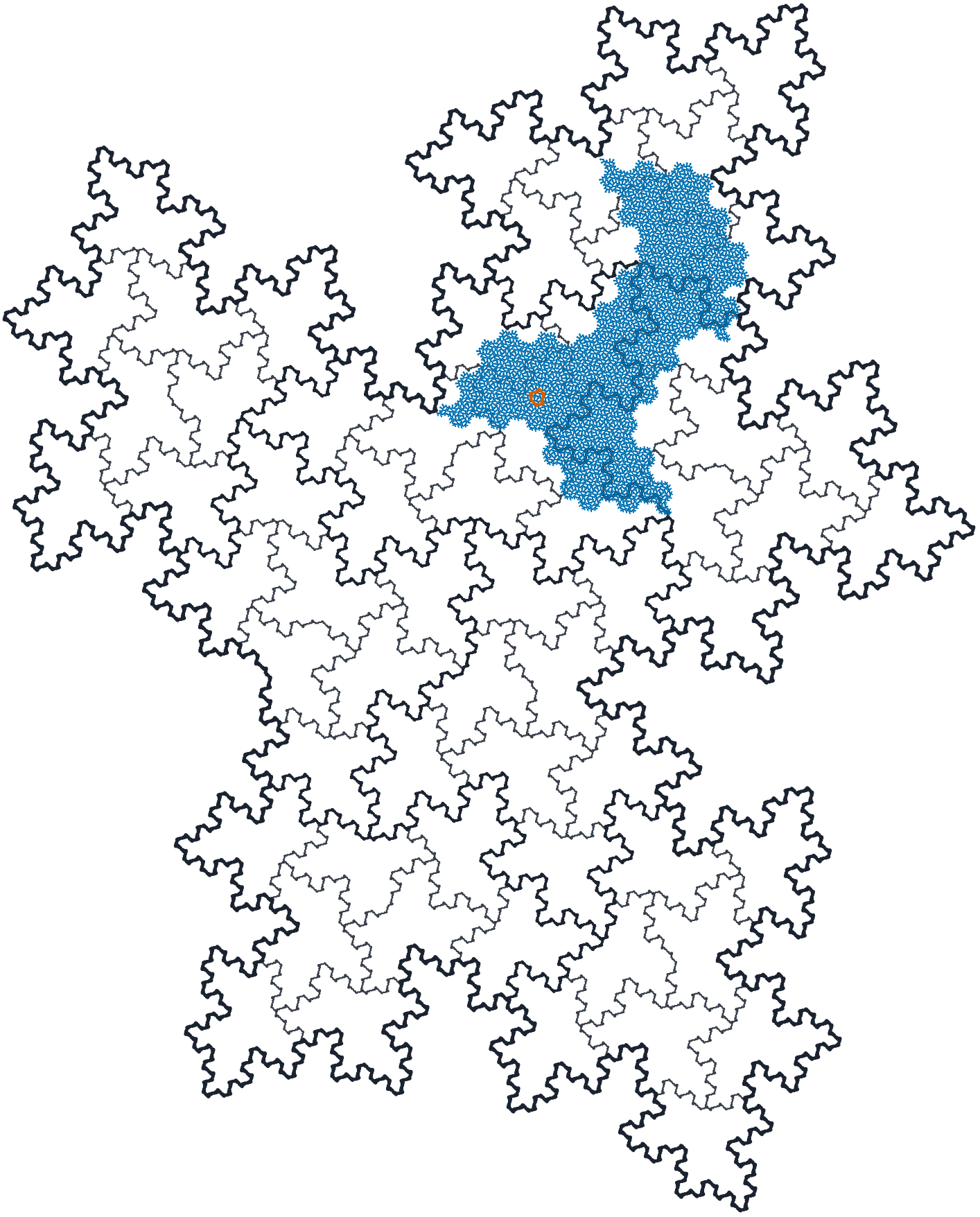
Inside the 71-tile block N_2. Every ribbon is drawn; the eight level-1 supertiles are outlined. The ribbons that pass through the orange block are highlighted: there are ten of them, so the 46 small pieces have already been gathered into ten longer ones. The nine blue ribbons are all parts of a single ribbon of the next block up; the orange one is part of another. Some of the blue ones are very short: they lie along the stretch of the block's edge that is also the rim of N_2, so they cannot be gathered yet.



Inside the 559-tile block N_3. Now only two ribbons pass through the orange block, and between them they carry all 46 of its pieces. The blue one wanders through 68 tiles before it leaves; the orange one through 69.



Inside the 4,401-tile block N_4. Still two ribbons. The blue one has grown to 687 tiles. The orange one is exactly the ribbon of the last picture, 69 tiles, because both its ends lie on the part of N_3's rim that is also the rim of N_4. The small block is the orange outline; the level-2 and level-3 supertiles are outlined, and the rest of the knot is in faint grey.



Inside the 34,649-tile block N_5. One ribbon does the whole job. It threads through 3,035 tiles and carries every one of the 46 pieces of the small block, the tiny orange outline. The level-3 and level-4 supertiles are outlined; the rest of the knot is left out.

Is the computed pairing really the knot in the pictures?

The program that draws the ribbons is separate from the program that follows ports up the levels. We took every supertile of levels 1, 2 and 3, all nine kinds, let the drawing program draw its ribbons, read off where each drawn piece starts and ends on the rim, and compared with the computed pairing. They agree on every one of the 1,516 ports of a level-3 Δ, and on every port of every other kind and level checked, with no rings. The drawing program draws the hand-made tile design exactly as it was drawn. The pictures in section 10 come from it too, and the counts they show (10, 2, 2, 1 ribbons through the small block) are the counts the port computation predicts.

Could a ring hide somewhere it is not counted?

A ring is finite, so some smallest supertile contains it. It is not inside a single child of that supertile (or a smaller one would contain it), so it must be found when that supertile's pairing is computed from its children: the walk visits every port and every leftover port is a ring. The count is zero at every assembly up to level 9, and the small-map walk, which also counts rings, finds none at any level above. A ring cannot hide inside a shared run either, because the zigzag always moves forward.

Does the zigzag really apply along every run?

It needs every corner along the run to be a hook on both sides. Corners along a run are, by definition, not marks, and every corner that is not a hook is a crossover corner, which is always a mark. It also needs every edge to use the slanted crossing, which is the one rule used at every edge of every tile.

Is the crossover always the same shape?

Four ribbon ends at two corners can be joined without hooks in two different ways. The small-map walk assumes the crossing one. We checked that this is the shape at every level and kind, and at the single tile.

Does the domino chain start correctly?

Levels 1 and 2 were computed outright, so the first two dominoes are pushed by hand. The knock-on step is only used from level 2 upwards. (For this design even the two-tile Γ cluster at the bottom of the tree has the signature, so the small-map walk also gives level 1 correctly, but the proof does not need that.)

Odd and even levels are different.

They are mirror images, because each level is built with a flip. Both maps were run through the walk, and the nest was checked starting at every level from 1 to 6, odd and even.

Does "the same map at every level" really hold for ever?

Yes. Section 8 proves it from Theorem 5.1 of the spectre paper, which says every level of the hierarchy is built by the same combinatorial rule: the lengths then obey one linear rule, so the small map, which only reads them as even or odd, must keep repeating once it has repeated once. We also checked it directly at every level from 2 to 9.

Does the nest really cover the whole plane?

Each block sits strictly inside its grandparent: they share no edge and no corner, and the smallest gap between them is at least two tile edges (measured at five levels, and part of the same repeating pattern). So the gap from the inner block to the rim of its ancestor k levels up grows without limit, and every point of the plane ends up inside some block. A supertile is a single solid region with one closed rim, so "inside" means what it should.

Are all the pieces of a block accounted for?

Every piece of the inner block gets a label: every hook and both crossover pieces. The computation reports both the number of ribbons the labels end up on and the number of labels lost to rings. The result is one ribbon and nothing lost, both in the exact version and in the small-map version.

Which tiling is this about?

The nest tiling: the one you get by growing the nest for ever. That is the tiling drawn on the main page. There are other spectre tilings, with a different chain of ancestors, or with two separate infinite chains meeting along a line. The same method applies to the first sort, but the merging has not been run for every possible chain. The second sort is not covered.

Is the design really the same on every tile?

Yes. The nest tiling uses only one handedness of the spectre (no mirror-image tiles), so "the same design" means exactly that: one drawing, turned and slid, never flipped.

What about over and under?

Separate from the question of one ribbon. Any ribbon drawing of this kind can be given an over-and-under weave that alternates along every ribbon: colour the regions between the ribbons like a chessboard and let the colour decide. The single-ribbon result is about which pieces join, not about the weave.

The proof in five lines

1. A supertile is a black box: all that matters is which rim ports its ribbons join.
2. For this design every box has the same signature: hooks everywhere, one crossover.
3. Hooked boxes glued along a run interact through even-or-odd alone (the zigzag).
4. So gluing is a finite walk on a small map, and the map is the same at every level: the signature passes from each level to the next like falling dominoes.
5. Carrying labels up a nest of supertiles that covers the plane shows every piece ends on one ribbon, and no ring ever forms.

Computed outright: every supertile of every kind, levels 1 to 9 (133 million tiles at level 9); the drawn ribbons against the computed pairing, levels 1 to 3; the nest from five starting levels exactly, and from six with the small map. Used from the spectre paper: every level of the hierarchy is built by the same combinatorial rule (Theorems 2.2 and 5.1); section 8 shows why that makes the small map repeat at every level.

The spectre: D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss, A *chiral aperiodic monotile*, Combinatorial Theory 4(2) (2024), arXiv:2305.17743. It followed the same authors' 'hat', *An aperiodic monotile*, Combinatorial Theory 4(1) (2024), arXiv:2303.10798. They credit Yoshiaki Araki, whose artistic explorations of the shape first prompted their study of it.

Design 0.1:01, drawn in the spectre tile editor. An earlier design, 0222221, was proved by the same method.